

Xinzhao An

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Expected December 2025

Master of Engineering in Financial Engineering

Relevant Coursework: Stochastic Calculus, Fixed Income Securities and Interest Rate Derivatives, Optimization Modeling in Finance, Machine Learning

University College London, London, UK

June 2024

Bachelor of Science in Mathematics with Statistical Science

Relevant Coursework: Probability and Inference, Further Linear Algebra, Stochastic Systems, Quantitative Modelling of Operational Risk and Insurance Analytics, Foundations of Machine Learning, Forecasting

SKILLS

Programming: Python (Pandas, NumPy, Matplotlib), Machine Learning (Scikit-learn, PyTorch), R, SAS, Excel

Certificate: FRM Level I (Passed)

RELEVANT EXPERIENCE

Pricing Analyst Intern, *RSA Insurance Group*, London, UK

June to Sept. 2023

- **Data Preprocessing:** Collected, cleaned, and structured six months of pet insurance data with ~100K records and 20+ variables; addressed missing values and standardized variables to enhance model robustness; handled missing or erroneous data with imputation or mean/mode replacement, adding error flags for tracking.
- **Pricing Model:** Utilized Generalized Linear Model (GLM) in Python to compute pet insurance quotes, selecting key risk factors based on significance testing; evaluated p-values of each factor and mitigated multicollinearity using techniques like variance inflation factors (VIFs); Visualized factor correlation analysis.
- **Workflow Automation:** Streamlined data importing, cleaning, transformation, and model execution processes in Python using libraries such as Pandas, NumPy, and scikit-learn. Developed reusable functions and scripts to automate data workflows, enabling consistent weekly updates with minimal manual intervention.

Quantitative Research Intern, *Shenwan Hongyuan Securities*

July to Aug. 2022

- **Multi-Factor Model:** Developed and executed a systematic stock selection model by ranking stocks based on factor exposures, optimizing portfolio composition through periodically rebalancing the portfolio; Evaluated each factor using Information Coefficient (IC) metrics to enhance accuracy and performance; Incorporated a dynamic stop-loss mechanism to manage risk.
- **Sector Rotation Strategies:** Developed and implemented a sector rotation strategy by analyzing the strength of financial and consumer sectors; Dynamically reallocating funds between sector ETFs based on relative performance metric.
- **Convertible Bond Pricing:** Used Black-Scholes formulas and derived PDE satisfied by convertible bond prices; Conducted comparative analysis with the Binomial Tree model to capture complex market dynamics; Evaluated discrepancies between theoretical and market prices, achieving an average pricing error of 6%.

PROJECTS

Dynamic Market Order Execution Strategy with Signal-Based Timing

Mar. to May 2025

- **Signal Construction & Evaluation:** Built real-time trading signals using MACD, RSI, EMA, and order flow imbalance; Conducted event studies around signal triggers to assess predictive power of each input.
- **Dynamic Execution Logic:** Designed rule-based and score-based algorithms that adjust aggressiveness within 60-second intervals; Tracked and optimized performance against TWAP benchmarks.

Mean-Reverting Portfolio Optimization via Semidefinite Programming

Feb. to May 2025

- **SDP-Based Portfolio Construction:** Formulated and solved a convex relaxation of the mean-reversion maximization problem using semidefinite programming in Python (CVXPY), incorporating volatility and sparsity constraints, as well as budget and leverage limits across multiple market sectors.
- **Strategy Implementation & Evaluation:** Applied Jurek and Yang's contrarian trading strategy to the optimized baskets; Benchmarked portfolio performance against the S&P 500 and equal-weighted portfolios, demonstrating improved returns.